Step 3

rain = none, maintenance = no, train = on time, appointment = attend

P(none) = 0.7

P(no | none) = 0.6

P(on time | none, no) = 0.9

P(attend | on time) = 0.9

$0.7 \times 0.6 \times 0.9 \times 0.9 = \boxed{0.3402}$ ← ans

Step 4

P(heavy, yes, delayed, miss) = $0.1 \times 0.1 \times 0.6 \times 0.4 = \boxed{0.0024}$

↑ ans

Step 6

P(rain) × [P(yes | rain) × P(delayed | rain, yes) + P(no | rain) × P(delayed | rain, no)]

none: $0.7 \times (0.4 \times 0.2 + 0.6 \times 0.1) = 0.098$

light: $0.2 \times (0.2 \times 0.4 + 0.8 \times 0.3) = 0.064$

heavy: $0.1 \times (0.1 \times 0.6 + 0.9 \times 0.5) = 0.051$

$0.098 + 0.051 + 0.064 = 0.213$

P(none | delayed) = 0.098 / 0.213 = 0.4601

P(light | delayed) = 0.064 / 0.213 = 0.3005

P(heavy | delayed) = 0.051 / 0.213 = 0.2394

Check:

0.4601 + 0.3005 + 0.2394 = 1

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Question 13:

because the code is doing the same computations I did by hand, summing over maintenance and dividing by the total. It's just called exact inference and it does it smarter.

Question 14:

Because appointment only depends on train and I know train is delayed so nothing to compute just read from the table

Question 16:

missing appointment is a clue the train was late, and a late train is a clue it rained, so the belief the rain goes up too, even if rain comes before everything else in chain.

Question 18:

Because when it's already raining heavy, a delay happens almost as often whether there's maintenance or not, so knowing about delay doesn't really tell us much extra about maintenance, and the number barely moves

ROCO

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